

Experiment No. 1

Indicator Phenolphthalein

N. of obs	Initial reading	Final reading	Volume of Acid used
1	0.0	8.4	8.4
2	8.4	16.9	8.5
3	16.9	25.3	8.4

mean value = $8.43 = V_1$

Indicator Methyl Orange

N. of obs	Initial reading	Final reading	Volume of acid used
	0.0	11.2	11.2
	11.2	22.2	11.0
	22.2	33.1	11.1

mean value = $11.1 = V_2$

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Experiment No. 1

Determination of a mixture of carbonate and hydroxide.

Introduction:-

Consider a mixture of NaOH and Na_2CO_3 .

Principle:-

Acid base titration

Chemicals and apparatus:-

Mixture of NaOH and Na_2CO_3 , deionized water, small beakers, burette, pipette, stand and burette clamp.

Indicator:-

Phenolphthalein and Methyl orange.

End point:-

Colorless and faint red.

Chemical equation:-



	$V_2 (\text{cm}^3)$	$V_1 (\text{cm}^3)$	$V_2 - V_1 = V_{\text{cm}^3}$
Volume of 0.1M HCl used (cm^3) = V_2 carbonate $= V_2 - V_1 = V_{\text{cm}^3}$	11.1	8.43	2.67
Volume of 0.1M HCl used (cm^3) $=$ all carbonate $= 2V_{\text{cm}^3}$	$2V = 2(2.67)$ $= 5.34$		
Volume of 0.10M HCl used (cm^3) $= \text{NaOH} = V_2 - 2V = V_{\text{cm}^3}$	$V_2 - 2V = 11.1 - 5.34$ $V_{\text{cm}^3} = 5.76$		

In case of Na_2CO_3

M HCl Na_2CO_3

$$M \times 2V = M \times 10$$

$$0.1 \times 5.34 = M \times 10$$

$$0.0534 = M$$

$$\text{Molarity of } \text{Na}_2\text{CO}_3 = 0.0534$$

$$\text{strength of } \text{Na}_2\text{CO}_3 = \text{Molarity} \times \text{Molar mass}$$

$$= 0.0534 \times 106$$

$$= 5.6604 \text{ g/dm}^3$$

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Theory:-

When a known volume of the mixture is titrated with HCl in the presence of phenolphthalein indicator, the acid reacts with all sodium hydroxide and with only half of the carbonate.

V_1 = all hydroxide + all carbonate

When a known volume of the mixture is titrated with HCl in presence of methyl orange indicator the acid reacts with all the hydroxide and all the carbonate

V_2 = all hydroxide + all carbonate

Volume of HCl = $\frac{1}{2}$ carbonate = $V_2 - V_1$

Procedure:-

i) Transfer with a pipette 10 cm³ of the mixture solution of

In case of NaOH

HCl NaOH

$$M \times (V_2 - 2V) = M \times 10$$

$$0.1 \times 5.76 = M \times 10$$

$$0.0576 = M$$

$$\text{Molarity of NaOH} = 0.0576$$

$$\text{Strength of NaOH} = \text{Molarity} \times \text{Molar mass}$$

$$= 0.0576 \times 40$$

$$= 2.304 \text{ g/dm}^3$$

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NaOH and Na_2CO_3 to a conical flask, and add one or two drops of phenolphthalein indicator.

iii) Add the acid from the burette till the solution becomes colorless.

iiii) Repeat the experiment twice or thrice and tabulate your result.

iv) Repeat the experiment with methyl orange until the color of the solution is changed to orange red.

vi) Repeat the experiment twice or thrice and tabulate your results.

vii) The volume of acid in the case is V_2 and is equivalent to all the hydroxide and all the carbonate.

viii) Calculate the strength of the sodium hydroxide and the sodium carbonate in the mixture.

No. of obs	Initial reading	Final reading Phenolphthalein	Final reading Methyl orange
1	0.0	8.4	11.1
2	11.1	16.8	22.1
3	22.1	25.2	33.0
Mean volume of HCl used (ccm ³)		8.4 = V_1	11.0 = V_2

	V_1 (cm ³)	V_2 (cm ³)	$(V_2 - V_1) \text{ cm}^3$
Volume of 0.1M HCl used (ccm ³)	8.4	11.0	2.6
$\frac{1}{2}$ carbonate = $V_2 - V_1 = V_{\text{cm}^3}$			

Volume of 0.1M

HCl used (ccm³)

= All carbonate = $2V_2$

$$= 2(2.6)$$

$$= 5.2$$

Experiment #2

Determination of a mixture of carbonate and hydride

Introduction:

Consider a mixture of NaOH and Na_2CO_3

Principle:

Acid base titration

Apparatus and chemicals:

Mixture of NaOH and Na_2CO_3 , deionized water, small beakers, burette, pipette stand and burette clamp.

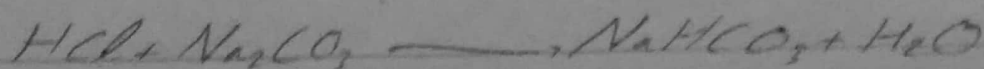
Indicator:

Phenolphthalein (ii), methyl orange

End point:

Colorless; faint red

Chemical equation:



In case of Na_2CO_3

HCl Na_2CO_3

$$M_1 \times 2V = M_2 \times 10$$

$$\frac{0.1 \times 5.2}{10} = M_2$$

$$M_2 = 0.052$$

Molarity of $\text{Na}_2\text{CO}_3 = 0.052$

Strength of $\text{Na}_2\text{CO}_3 = M_1 \times 106$

$$= 0.052 \times 106$$

$$= 5.72 \text{ g/dm}^3$$

In case of NaOH

HCl NaOH

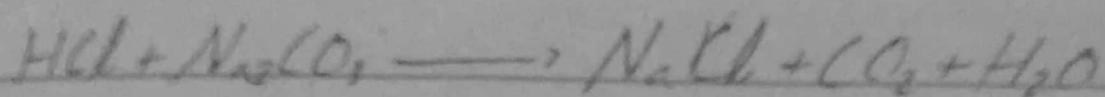
$$0.1 \times 27 = M_2 \times 10$$

$$M_2 = 0.027$$

Strength of $\text{NaOH} = 0.027 \times 40$

$$= 1.08 \text{ g/dm}^3$$

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Theory:-

When a known volume of the mixture is titrated with HCl in presence of phenolphthalein indicator. The acid reacts with all the sodium hydroxide and with only half of the carbonate

$$V_1 = \text{all hydroxide} + \frac{1}{2} \text{ the carbonate}$$

When a known volume of mixture is treated with HCl in presence of methyl orange the acid reacts with all hydroxide and all the carbonate

$$V_2 = \text{all hydroxide} + \text{all carbonate}$$

$$\text{Volume of HCl} - \frac{1}{2} \text{ carbonate} = V_2 - V_1 = V_{\text{NaOH}}$$

$$\text{Volume of HCl} = \text{all carbonate} = 2V_{\text{NaOH}}$$

$$\text{Volume of HCl} - \text{NaOH} = V_2 - 2V_{\text{NaOH}}$$

Procedure:-

- i) Transfer with a pipette 10 cm^3 of the mixture of the solution of NaOH and Na_2CO_3 to a conical flask, and add one or two drops of phenolphthalein indicator.
- ii) Add the acid from the burette till the solution become colorless.
- iii) The volume of acid in the case is V_1 and is equivalent to all hydroxide and half the carbonate.
- iv) Then add two drops of methyl orange to the same flask and continue titration to faint red.
- v) The volume of the acid in the case is V_2 and is equivalent to the second half of the carbonate.

vii Repeat the experiment three times and tabulate your results as.

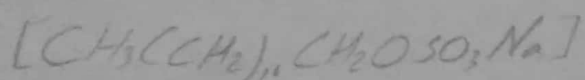
viii Calculate the strength of sodium hydroxide and sodium carbonate in the mixture.

Experiment No 3

Composition

Precipitated chalk = 57.0 gram CaCO_3

Sodium lauryl sulfate = 1.0 gram



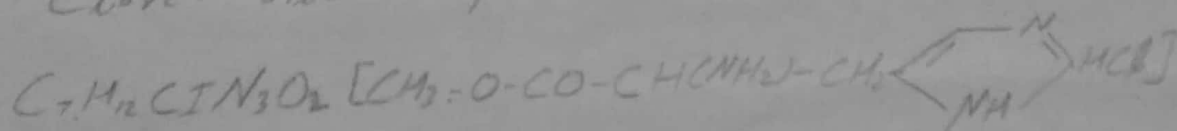
Glycerin = 21.0 gram $\text{C}_3\text{H}_8\text{O}_3$

Starch = 3.0 gram $(\text{C}_6\text{H}_{10}\text{O}_5)_n$

Water = 19.5 cm³ H_2O

Saccharine = 1 drop

Clove. oil = 1 drop



Preservative = 1.0 spatula $\text{C}_6\text{H}_5\text{COONa}$

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Experiment #3

Preparation of sample of tooth paste and examine its properties

Theory:-

Dentifrices are the agents used along with a tooth brush to clean, maintain the health and polish natural teeth. They are used in paste, powder, gel or liquid form.

Reason for using Dentifrices:-

Dentifrice
minimize the build up of plaque
strengthening teeth against caries
(tooth decay), removing staining, removing
of food debris, re-moisturizing dry
mouth, alleviating (to reduce pain) sensitivity
and combating or preventing
halitosis (bad breath).

Types:-

- i) Anti plaque tooth paste
 - ii) Fluoride tooth paste.
 - iii) Whitening tooth paste
 - iv) Sensitive tooth tooth paste etc.
- Working of toothpaste.

Toothpaste contains abrasive ingredients that scrub away stains and plaque from teeth. Preservatives retard the growth of bacteria.

Procedure:-

21 g of glycerine and starch is added in a beaker with constant stirring. The 145 ml of water and 1 gram of sodium lauryl sulfate is mixed in another beaker. Then solutions of both beakers are mixed together. 57.5 gram of glycerine finally

divided bulk is then added and stirring the whole mixture to get homogenize paste. One or two drops of clove oil are mixed while stirring. One drop of saccharine as sweetner is added in the mixture while continuously stirring to homogenize the whole mixture. At the end 1% spatula of preservative is added. Stirring is continued till homogenize paste is obtained.

Characteristics of tooth paste:-

pH = 7.8

Consistency = uniform and homogenize

Color = white

Result:-

A sample of tooth prepared.

Observation and Calculation

Weight of empty beaker = 46g

Weight of beaker + polymer formed = 54g

Weight of polymer formed = $54 - 46$
= 8g

Result -

Weight of polymer is 8g



Experiment #4

Prepare Urea formaldehyde (U-F) resin.

Theory:-

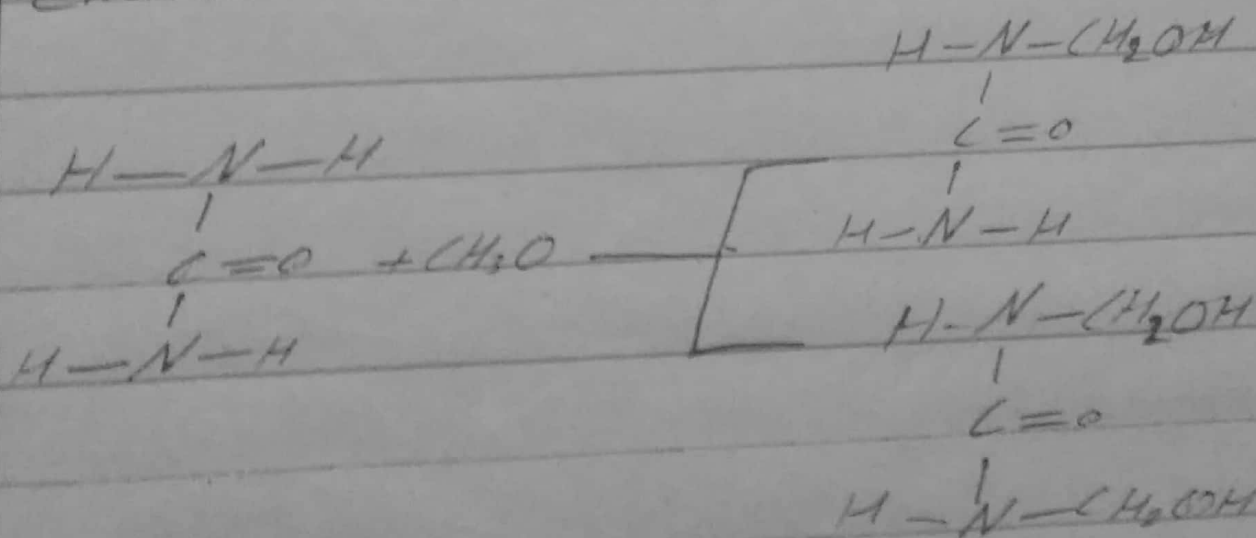
Urea formaldehyde resins are formed by condensation of urea and formaldehyde in acidic medium.

Chemicals and Apparatus:-

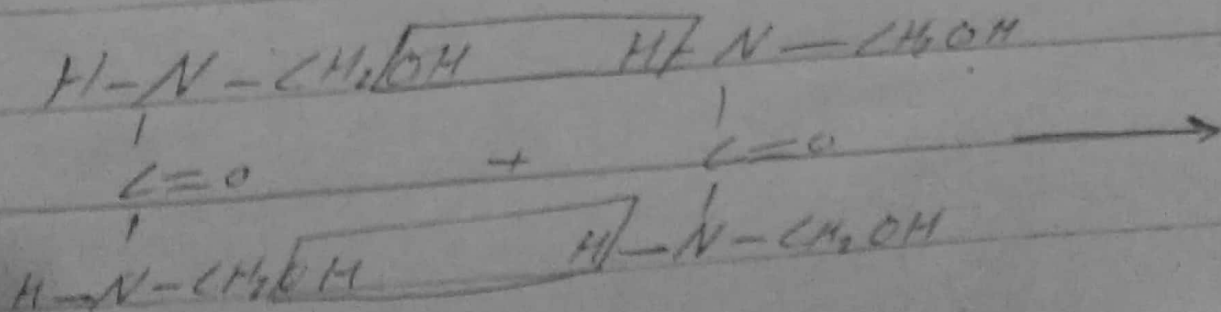
Urea (2g), 40% aqueous

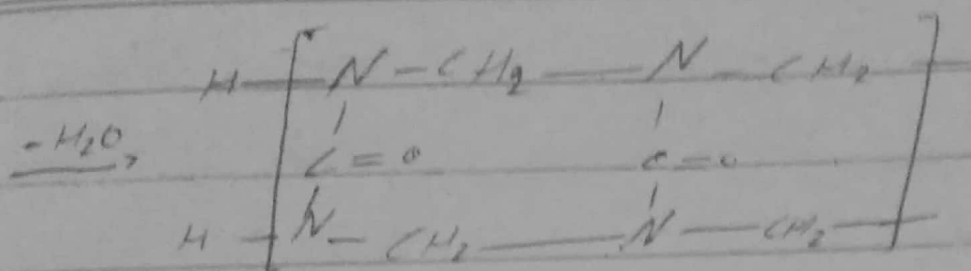
formaldehyde solution or formalin (5ml)
conc H_2SO_4 , watch glass, measuring cylinder.

Chemical reactions:-



N,N-Dimethyl urea





Procedure:-

Take 5ml of formalin in a beaker. Add 2g urea powder stir with a glass rod to make a saturated solution. Add a few drops of conc H_2SO_4 and stir vigorously till a white solid mass is formed. Filter the residue and wash it several times with distilled water to remove any acid. Dry the residue into folds of filter paper or in an oven and weigh. Report the yield of urea formaldehyde formed.

Result:-

Weight of urea formaldehyde resin is g.

Experiment No. 5

Observation and Calculation

Indicator phenolphthalein

No. of obs	Initial reading	Final reading	Volume of HCl used (cm ³)
1	0.0	6.3	6.3
2	6.3	12.7	6.4
3	12.7	19.0	6.3

$$\text{Mean} = 6.33 = V_1$$

Indicator Methyl Orange

No. of obs	Initial reading	Final reading	Volume of HCl used
1	0.0	16.5	16.5
2	16.5	32.9	16.4
3	32.9	49.4	16.5

$$\text{Mean} = 16.46 \text{ cm}^3 = V_2$$

Experiment No. 5

Determine a mixture of carbonate and bicarbonate

Introduction:-

Consider a mixture of Na_2CO_3 and NaHCO_3 .

Principle:-

Acid base titration

Apparatus and chemicals:-

Mixture of Na_2CO_3 and NaHCO_3 , deionized water
small beakers, burette, pipette, stand
and burette clamp.

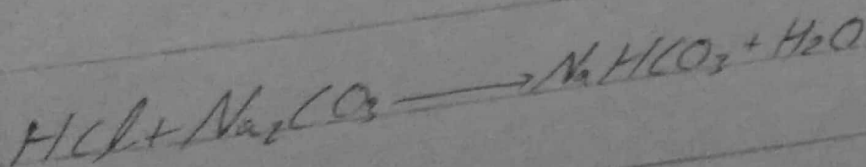
Indicator:-

Phenolphthalein is, methyl
orange

End Point:-

Reddish, colorless

Chemical equation:-



$$\text{Volume of } 0.1 \text{ HCl used} = \frac{1}{2} \text{ carbonate} \\ = 6.33 = V_1$$

$$\text{Volume of } 0.1 \text{ HCl used all carbonate} = 12.66 = 2V_1$$

$$\text{Volume of } 0.1 \text{ M HCl used} = \text{all carbonate} + \text{all bicarbonate} \\ = 16.46$$

$$\begin{aligned} \text{Volume of } 0.1 \text{ M HCl used} &= \text{NaHCO}_3 \\ &= V_2 - 2V_1 \\ &= 16.46 - 12.66 \\ &= 3.8 \text{ cm}^3 \end{aligned}$$

In case of Na_2CO_3



$$0.1 \times 12.66 = M/2 \times 10$$

$$0.063 = M$$

Strength of $\text{Na}_2\text{CO}_3 = \text{Molarity} \times \text{Molar mass}$

$$= 0.063 \times 106$$

$$= 6.678 \text{ g/litre}$$



Theory:-

When a known volume of mixture is titrated with HCl in presence of phenolphthalein indicator, the acid reacts with only half of the carbonate

$$V_1 = \frac{1}{2} \text{ the carbonate}$$

When a known volume of the mixture is titrated with HCl in presence of methyl Orange indicator, the acid reacts with all the carbonate and all the bicarbonate

$$V_2 = \text{all carbonate} + \text{all bicarbonate}$$

$$\text{Volume of HCl} = \frac{1}{2} \text{ carbonate} = V_1 \text{ cm}^3$$

$$V_2 \text{ cm}^3 = \text{Volume of HCl} = \text{all carbonate} + \text{all bicarbonate}$$

$$\text{Volume of HCl} = \text{NaHCO}_3 = (V_2 - 2V_1) \text{ cm}^3$$

Procedure:-

Transfer with a pipette 10 cm^3 of the mixture solution of

In case of NaHCO_3

HCl NaHCO_3

$$0.1 \times 3.8 = 11.84$$

$$M = 0.038$$

Strength of $\text{NaHCO}_3 = 11.84$

$$= 0.038 \times 84$$

$$= 3.192$$

NaHCO_3 and Na_2CO_3 to a conical flask, and add one or two drops of phenolphthalein indicator.

Add the acid from the burette till the solution becomes colorless.

Repeat the experiment twice or thrice and tabulate your results.

The volume of acid in case is V_1 and is equivalent to half the carbonate.

Repeat the experiment with methyl orange until the color changes to faint red.

Repeat the experiment twice or thrice and tabulate your results.

The volume of acid in the case is V_2 and is equivalent to

all the carbonate and bicarbonate.

Calculate the strength of sodium bicarbonate and sodium carbonate.

in the mixture.

No of con	Initial reading	Final phenolphthalein reading	Final reading Methyl orange
1	0.0	6.5	16.5
2	0.0	6.6	16.4
3	0.0	6.4	16.5
Volume of 0.1M HCl used (Mean)		$6.43 = V_1$	$16.46 = V_2$

Volume of 0.1M HCl used all carbonate
 $= 2V_1$

$$= 2 \times 6.43$$

$$= 12.86$$

Volume of 0.1M HCl used all bicarbonate
 $= V_2 - V_1$
 $= 10.03$

Experiment No. 6

Determine a mixture of carbonate and bicarbonate

Introduction:-

Consider a mixture of Na_2CO_3 and NaHCO_3

Principle:-

Acid base titration

Apparatus and chemicals:-

Mixture of Na_2CO_3 and NaHCO_3 , deionized water, beakers, burette, pipette, stand and burette clamp

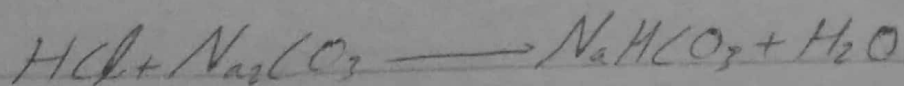
Indicator

Phenolphthalein, Methyl orange

End point

Colorless, just red

Chemical equations:-



In case of Na_2CO_3

HCl Na_2CO_3

$$0.1 \times 2V_1 = M/2 \times 10$$

$$0.1 \times 12.86 = \frac{M}{2} \times 10$$

$$M = 0.063$$

$$\begin{aligned}\text{Strength of } \text{Na}_2\text{CO}_3 &= 0.063 \times 106 \\ &= 6.678 \text{ g/lm}^3\end{aligned}$$

In case of NaHCO_3

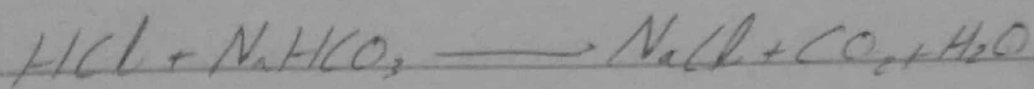
HCl NaHCO_3

$$0.1 \times (V_2 - V_1) = M \times 10$$

$$0.1 \times 10.23 = M \times 10$$

$$M = 0.1023$$

$$\begin{aligned}\text{Strength of } \text{Na}_2\text{CO}_3 &= 0.1023 \times 84 \\ &= 8.42 \text{ g/lm}^3\end{aligned}$$



Theory:-

When a known volume of the mixture is titrated with HCl in presence of phenolphthalein indicator, the acid reacts with only half of the carbonate.

$$V_1 = \frac{1}{2} \text{ carbonate}$$

When a known volume of the mixture is titrated with HCl in presence of methyl orange indicator, the acid reacts with all the carbonate and all bicarbonate.

$$V_2 = \text{all carbonate} + \text{all bicarbonate}$$

$$\text{Volume of HCl} = \frac{1}{2} \text{ carbonate} = V_1 \text{ cm}^3$$

$$\text{Volume of HCl} = \text{all carbonate} + \text{all bicarbonate} = V_2$$

$$\text{Volume of HCl} = \text{NaHCO}_3 = (V_2 - 2V_1) \text{ cm}^3$$

Procedure:-

Transfer with a pipette 10 cm³ of the mixture solution of NaHCO₃

and Na_2CO_3 to a conical flask
and add one or two drops of
phenolphthalein indicator

Add the acid from the burette
till the solution becomes colorless.

The volume of acid in the case
is V_1 and is equivalent to
half the carbonate.

Then add two drops of methyl
orange to the same conical flask
and continue titration to the
faint red color or the orange
color.

The volume of acid in the case
is V_2 and is equivalent to the
second half of the carbonate
and all the bicarbonate.

Repeat the experiment three
times and tabulate your results.

Calculate the strength of the
sodium carbonate and sodium bicarbonate
in the mixture.